

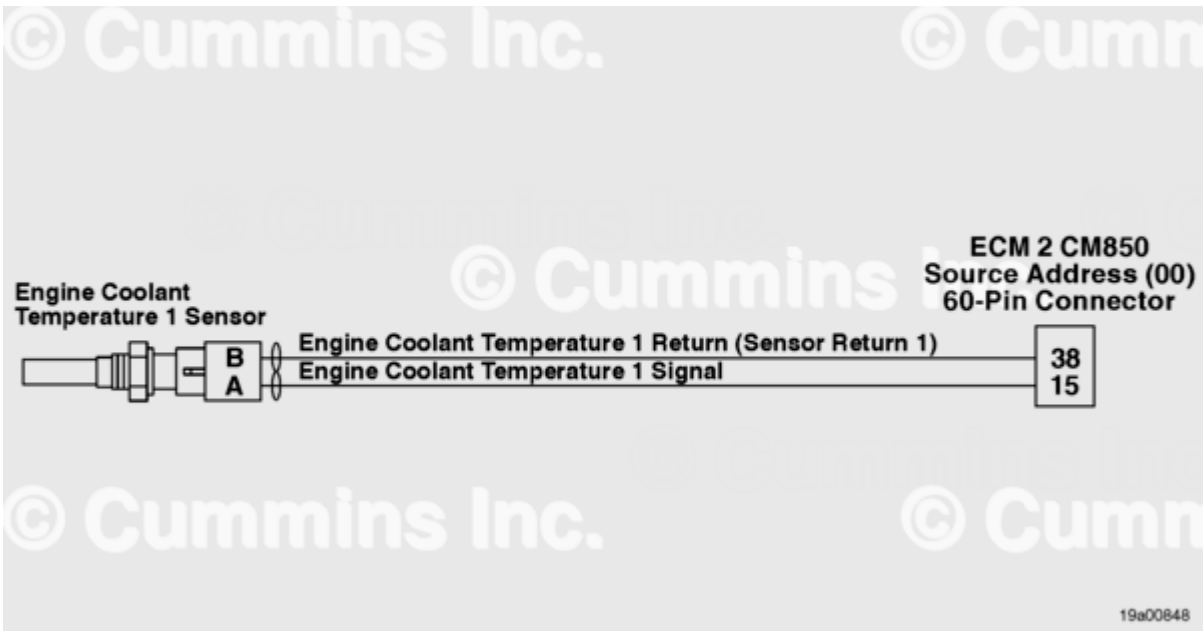
Fault Code: 151

Engine Coolant Temperature - Data Valid But Above Normal Operating Range - Most Severe Level

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 151 PID(P): P110 SPN: 110 FMI: 0/0 Lamp: Red SRT:	Engine coolant temperature signal indicates engine coolant temperature above engine protection critical limit.	Engine will shut down.



Engine Coolant Temperature Sensor Circuit

Circuit Description

The engine coolant temperature sensor is a variable resistor sensor used by the ECM to monitor the engine coolant temperature. The engine coolant temperature sensor has two circuits: signal, and return circuits. The signal voltage indicates the coolant temperature.

Component Location

The engine coolant temperature sensor is located on the thermostat housing.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the generator set controller is active or when the engine is running.

Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected the engine coolant temperature was greater than the engine protection limit.

Action Taken When The Fault Code Is Active

- The generator set controller displays a shutdown fault immediately when the diagnostics runs and fails.
- The torque output of the engine will be reduced.
- The engine will be shut down if the Engine Protection Shutdown feature is enabled.

Conditions For Clearing The Fault Code

- To validate the repair, bring the engine up to operating temperature and run it in normal loaded conditions for 15 minutes.
- The generator set controller will turn off the shutdown indicator immediately after the user presses the reset.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

Shoptalk

Possible causes of this fault code include:

- Obstructed airflow through the radiator.
- Engine coolant level low
- Collapsed or restricted cooling system hoses.
- Malfunctioning engine coolant thermostat.
- A malfunctioning water pump.
- Malfunctioning engine coolant temperature sensor

Refer to Troubleshooting Fault Code 151 . (</qs3/pubsys2/xml/en/procedures/60/60-t05-151.html>)

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